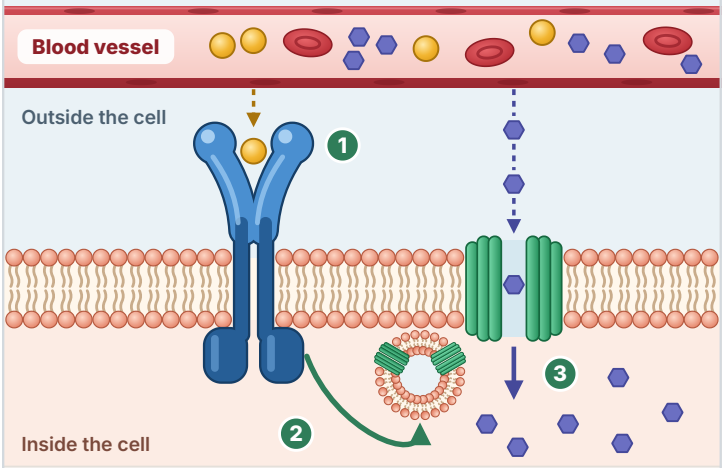
Patient name _____ Date _____

Understanding Insulin Resistance

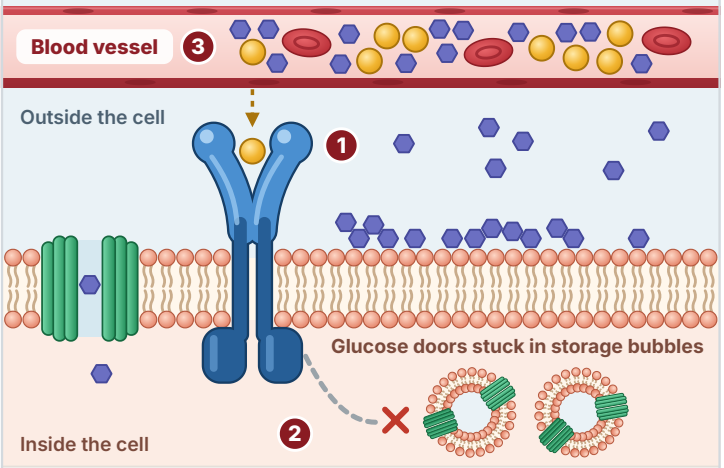
Insulin is a hormone your pancreas releases, especially after you eat. It tells your muscle and fat cells to open their **glucose doors** and your liver to store sugar (glucose), so it leaves your blood and fuels your body. With **insulin resistance**, the cells stop listening well. Sugar starts to back up, so the pancreas pushes out more and more insulin to keep blood sugar normal. This can go on quietly for years before blood sugar climbs high enough to be called prediabetes or type 2 diabetes.

NORMAL — cells answer insulin



1. Insulin from the blood (gold dashes) fits into its receptor.
2. The signal (green arrow) lifts a bubble of doors to the surface.
3. Glucose (purple arrows) goes in through an open door.

INSULIN RESISTANCE — cells barely answer



1. Insulin still fits its receptor, but the cell barely listens.
2. The signal (gray dashes) dies at the X. Most doors stay inside.
3. Glucose stays in the blood. The pancreas makes more insulin.

KEY Glucose (sugar) | Insulin | Insulin receptor | GLUT4 glucose door | Storage bubble | Cell membrane | Red blood cell

WHAT DRIVES IT

Sugary drinks and processed food
Sitting most of the day
Short sleep, long-term stress
Extra fat around the belly
Family history, age 45 or older
Some medicines, such as steroids

CLUES TO WATCH FOR

Often no signs at all
Blood sugar creeping up on labs
High fasting insulin on a lab test
Dark skin at the neck or armpits
Skin tags in those same spots
High triglycerides, low good HDL

WHERE IT CAN LEAD

Prediabetes, then type 2 diabetes
Heart disease and stroke
High blood pressure
Fatty liver disease
PCOS and hormone shifts
Memory and thinking problems

MORE THAN 1 in 3 U.S. adults have prediabetes, and **more than 8 in 10** of them do not know it (CDC). Insulin often climbs years before blood sugar does, so ask your doctor about a **fasting insulin** test along with the numbers below.

Know your numbers

Review your results with your pharmacist. **Fasting insulin** _____ µIU/mL **Lab range** _____

Fasting blood sugar mg/dL, after 8 hours with no food

NORMAL	PREDIABETES	DIABETES	My result
under 100	100 to 125	126 or more	

A1c average blood sugar over about 3 months

NORMAL	PREDIABETES	DIABETES	My result
under 5.7%	5.7 to 6.4%	6.5% or more	

Metabolic syndrome check Check each sign you have. If you take medicine for one, check it. **Any 3 of these 5** means metabolic syndrome.

Waist	Triglycerides	HDL (good) cholesterol	Blood pressure	Fasting blood sugar
☐	☐	☐	☐	☐
40 inches or more (men) 35 inches or more (women)	150 mg/dL or more	under 40 mg/dL (men) under 50 mg/dL (women)	130/85 or higher (either number)	100 mg/dL or more

Sources: American Diabetes Association (a diabetes result is confirmed with a repeat test); NHLBI and American Heart Association. Cell drawing not to scale.

Lifestyle-Based Medicine

When you eat, what you eat and how you move: three levers that help your cells answer insulin

10-hour eating window

- Example: 8 AM to 6 PM
- Same hours, weekends too
- Outside it: water, plain tea or black coffee
- On blood sugar medicine? Ask your doctor first

A slow-sugar plate

- Non-starchy veggies: half the plate
- Veggies and protein first
- Slow carbs last, about a fist
- Whole fruit, not juice

Move often, sleep well

- Walk 10 minutes after meals
- Strength training 2 to 3 days a week
- Stand up every 30 minutes
- Sleep 7 to 9 hours a night

Nutrient Solutions

What your body needs, and the NutriDyn formula that delivers it

START HERE

All-in-one glucose support

dihydroberberine, chromium, alpha-lipoic acid



Gluco IR

150 capsules · 30 servings

- Promotes healthy blood glucose levels already in the normal range ◆
- Supports the normal action of insulin (chromium) ◆

IN EACH SERVING

Per 5 capsules: 100 mg dihydroberberine, 300 mg alpha-lipoic acid, 600 mcg chromium, 2,000 mcg biotin, 500 mg Ceylon cinnamon, 200 mg gymnema, plus fenugreek, nettle, bitter melon, green tea and banaba

5 capsules daily
with meals

Cell energy switch

dihydroberberine + cinnamon



Berberine Pro

90 capsules

- Promotes healthy glucose metabolism ◆
- Supports heart health and healthy lipid balance ◆

IN EACH SERVING

Per capsule: 100 mg dihydroberberine + 84 mg organic Ceylon cinnamon bark extract

1 capsule 3 times daily
before meals

Friendly gut bacteria

Akkermansia + 4 probiotic strains



UltraBiotic Akkermansia Plus

30 capsules

- Supports healthy glucose metabolism ◆
- Supports healthy weight and a healthy gut lining ◆

IN EACH SERVING

Per capsule: 500 million TFU Akkermansia muciniphila (80% pasteurized), 22 billion CFU B. animalis and L. rhamnosus GG, 30 million CFU C. butyricum

1 capsule daily
any time, such as bedtime

Essential omega-3 fats

EPA + DHA from fish oil



Omega Pure EPA-DHA 1000

120 softgels

- Supports healthy cardiovascular function ◆
- Supports healthy cell membranes in the body ◆

IN EACH SERVING

Per softgel: 1,000 mg EPA + DHA (730 mg EPA, 270 mg DHA) from anchovy, sardine and mackerel

1 softgel twice daily
with breakfast and dinner

YOUR DAILY RHYTHM

Morning

Gluco IR
3 capsules
with breakfast
Berberine Pro
2 capsules:
1 before breakfast,
1 before lunch
Omega Pure
EPA-DHA 1000
1 softgel with
breakfast

Evening meal

Gluco IR
2 capsules
Berberine Pro
1 capsule
before dinner
Omega Pure
EPA-DHA 1000
1 softgel

Bedtime

UltraBiotic
Akkermansia Plus
1 capsule

The full plan is **11 pills a day**. Overwhelmed? Start with the **Start here** pick, then add one product a week.

PHARMACIST CHECK

Keep taking your medicines.

- **Both have dihydroberberine.** Gluco IR and Berberine Pro give 400 mg a day together: avoid both if pregnant or breastfeeding. It can slow the enzymes (CYP3A4, 2D6, 2C9) that clear drugs such as warfarin and cyclosporine. Fish oil and fenugreek can add to bleeding on a blood thinner: bring your full list.
- **Insulin or a sulfonylurea** (such as glipizide)? Ask your prescriber before you start, and watch for low blood sugar. **Metformin** can lower B12 over time: ask for a check.
- **Thyroid medicine?** Take it several hours apart from Gluco IR (with a morning thyroid pill, move Gluco IR to lunch and dinner): its chromium can lower how much you absorb. Gluco IR's biotin can throw off some lab tests: tell the lab.
- **Weak immune system** or on cyclosporine? Ask your pharmacist before Akkermansia Plus: it has live bacteria.
- **Allergic to fish, peanuts or chickpeas?** Ask your pharmacist first: Omega Pure is fish oil, and the fenugreek in Gluco IR can trigger it too.

Special considerations for you:

Recommended by _____ Phone _____

◆ These statements have not been evaluated by the Food and Drug Administration. These products are not intended to diagnose, treat, cure, or prevent any disease.

NutriDyn